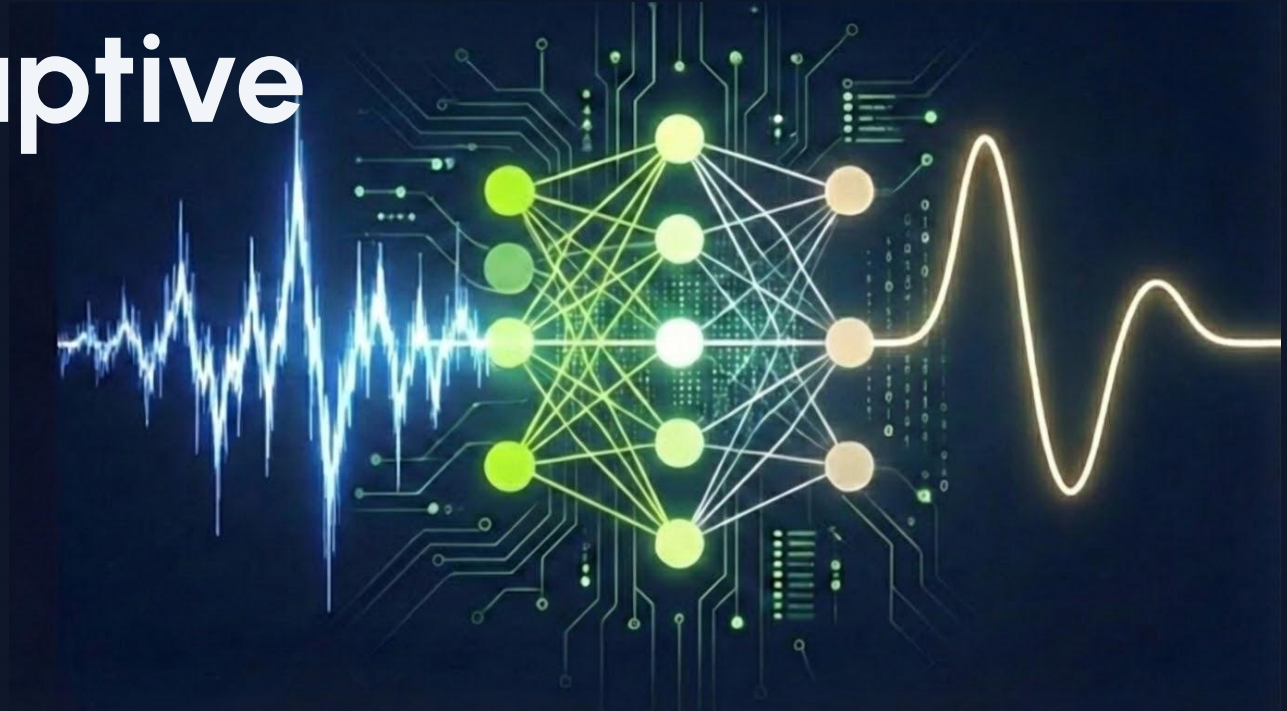


Intelligent Partial Discharge **Denoising** with Adaptive Capabilities

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University of Malaya | 2026



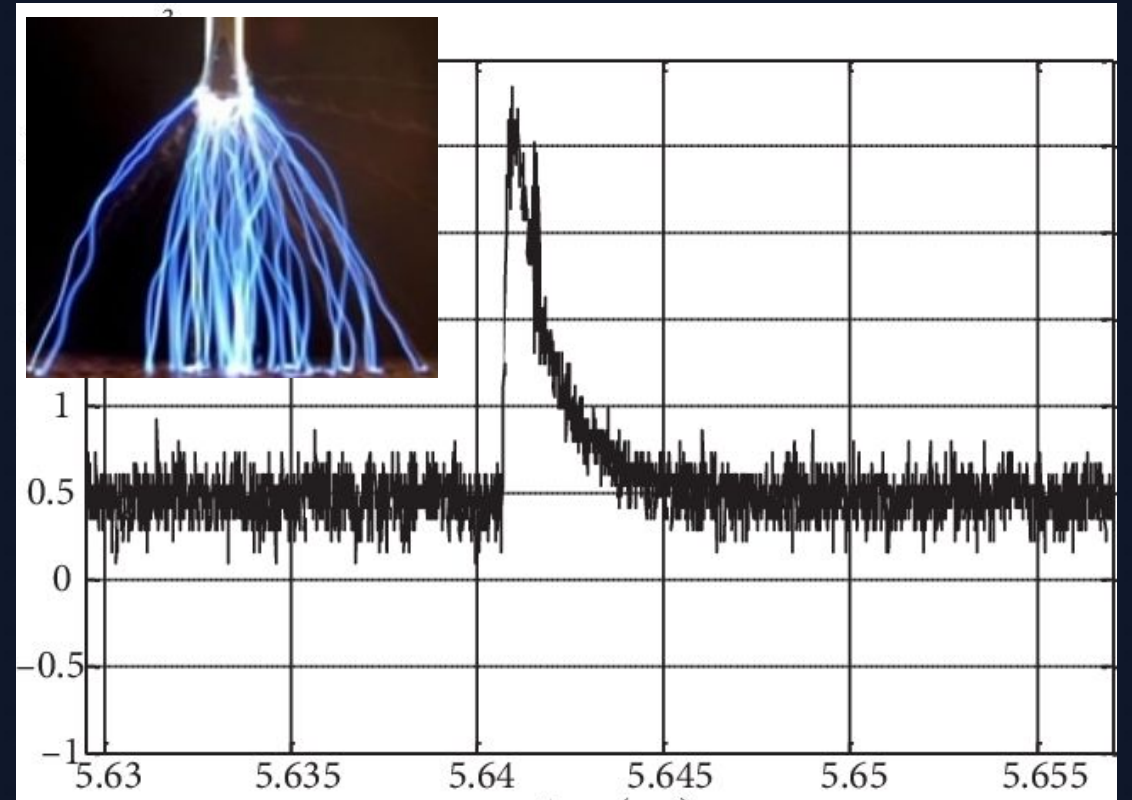
The Challenge: Signal vs. Noise

What is Partial Discharge (PD)?

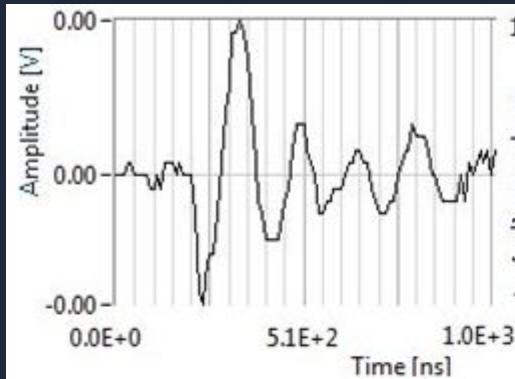
Partial discharge is a localized electrical dielectric breakdown of a small portion of a solid or fluid electrical insulation system under high voltage stress.

Why it Matters?

Accurate diagnostic analysis of PD signals is the critical cornerstone for condition monitoring of high-voltage switchgear. It serves as an early warning system for insulation failure.

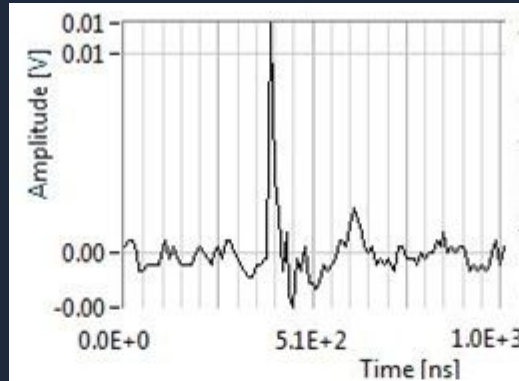


Types of Partial Discharge



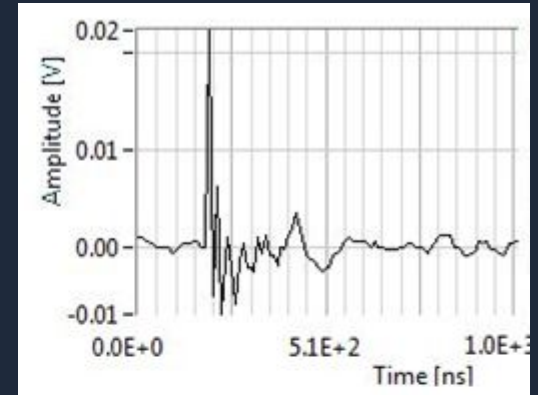
Internal Discharge

Occurs within voids or cavities in solid insulation. Characterized by short rise times and high amplitudes (MHz range).



Surface Discharge

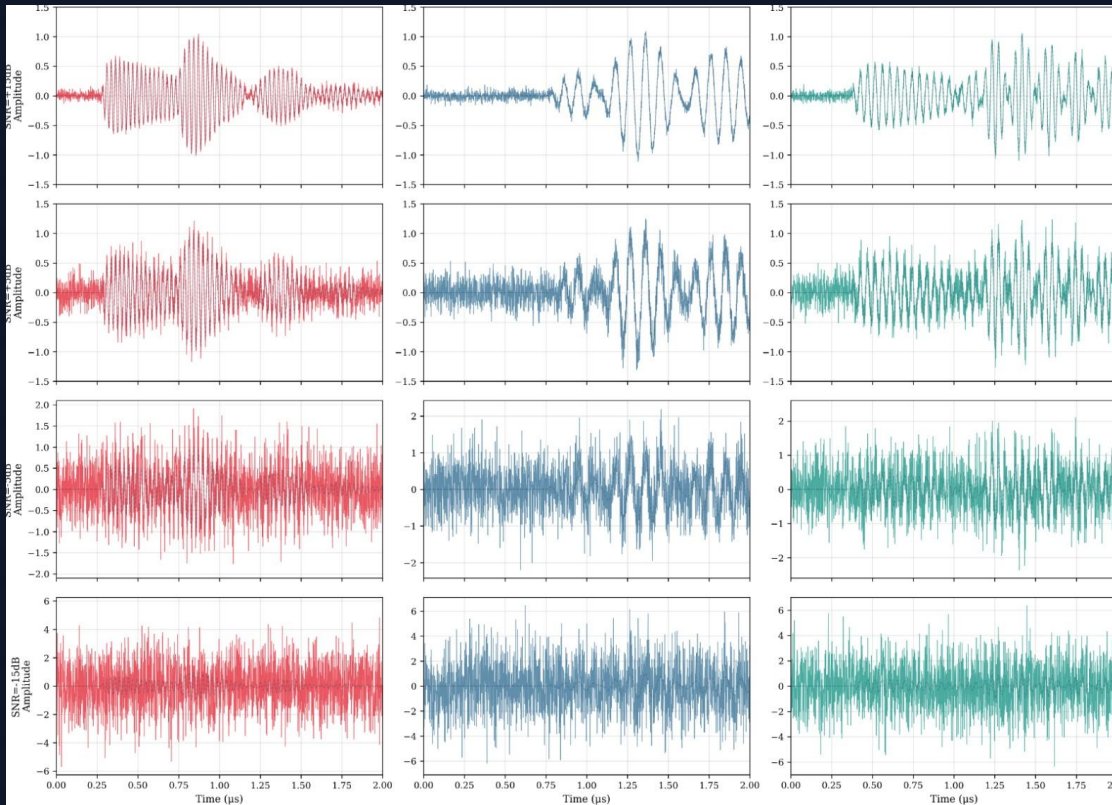
Develops along insulation boundaries. Signals have broader pulse widths and irregular occurrence patterns.



Corona Discharge

Occurs at sharp edges in gas-insulated systems. Repetitive pulses strongly correlated with voltage polarity.

Noise Contamination in Real-World Setting



The Signal Buried in Noise

On-site measurements are compromised by a complex mixture of non-stationary noise. In extreme conditions (-20 dB SNR), the PD signal is completely obscured.

- ⚡ **White Noise:** Random thermal agitation.
- ⚡ **Narrowband Interference:** Discrete Frequency Noise.
- ⚡ **Impulse Noise:** Stochastic switching transients.
- ⚡ **Other Powerline Noise:** 50/60 Hz Interference.

Why Traditional Methods Fail

STATIC AND MANUAL DEPENDENCY

Wavelet and filter-based methods rely on fixed mathematical bases. They require manual selection of the mother wavelet and thresholding parameters, which cannot adapt to changing noise conditions.

MORPHOLOGICAL DISTORTION

Traditional Method like Soft Thresholding, Savitzky-Golay often decapitates the signal peaks, reducing amplitude and altering rise-time. This distortion can lead to misclassification of the defect.

Poor Stopband Suppression

High-frequency noise components often leak through the filter, remaining in the output signal and corrupting the diagnostic quality, specifically when calculating derivatives of the data.

Boundary and Edge Artifacts

When this window reaches the start or end of the data stream, the filter lacks sufficient neighboring data points to compute the polynomial correctly. This results in ringing effect at the edges of the signal,

Data Acquisition Strategy

Synthetic Data Generation

Mathematical modeling of PD pulses using single and double damped oscillation formulas.

$$D_1 = A_1 e^{-\frac{t}{\tau}} \sin(2\pi f_c t) \text{ for } t \geq t_0$$

$$D_2 = A_2 [e^{-\frac{1.3t}{\tau}} - e^{-\frac{2.2t}{\tau}}] \sin(2\pi f_c t) \text{ for } t \geq t_0$$

Real-World Augmentation

To address data scarcity, **Wasserstein Generative Adversarial Network (WGAN)** was employed.

The WGAN learns the latent statistical distributions of real on-site PD pulse and synthesizes high-fidelity synthetic pulses, creating a realistic hybrid dataset.

Real World Data Source

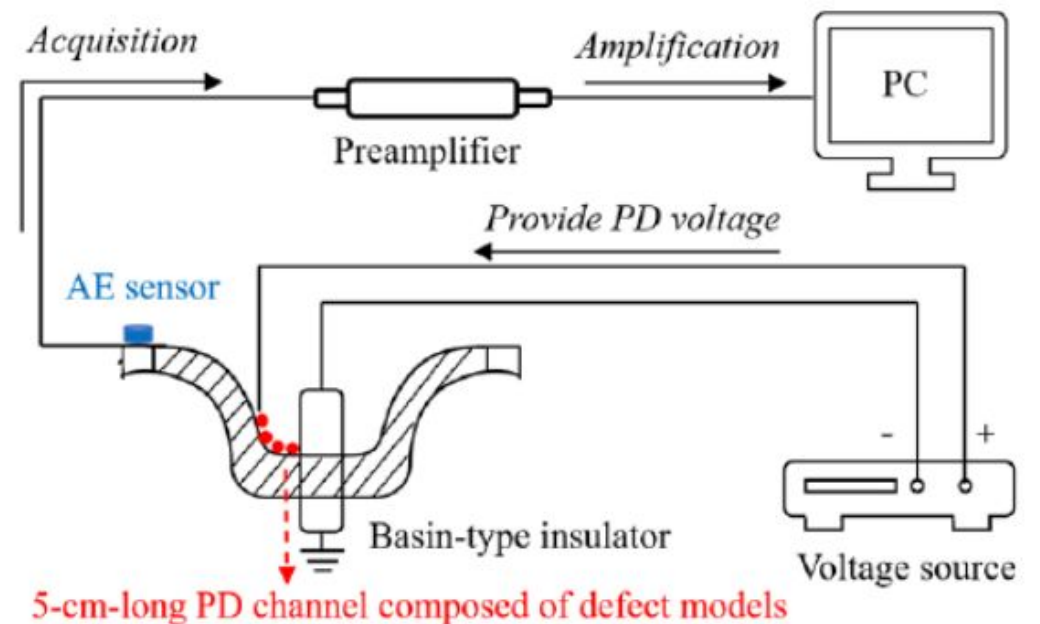
Real World Data Acquisition

Source: Lin et al. (2022) Dataset.

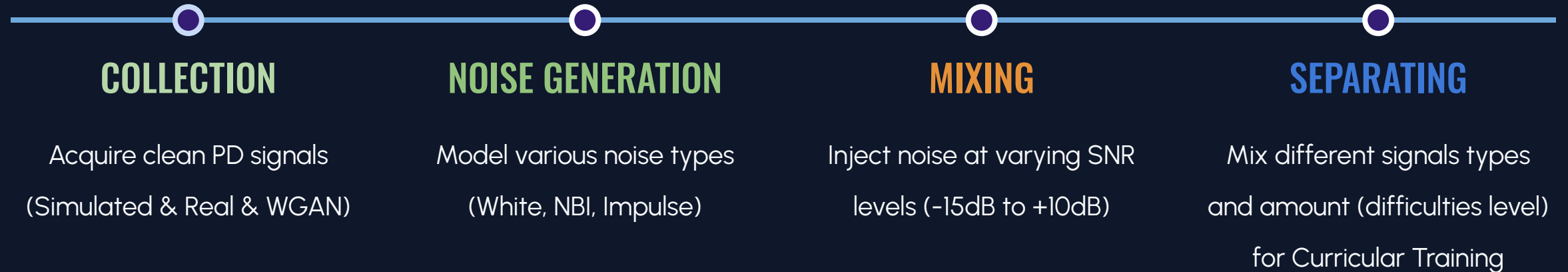
Setup: 15kV HV source, Piezoelectric AE sensors.

Defects: Copper beads on insulating structures (simulating surface/internal discharge).

Role: Used as the 'Ground Truth' for real noise characteristics and final validation testing.



Data Acquisition Strategy



The Proposed Solution:

MR-TAE

Multi-Level Wavelet

Recurrent-Transformer Autoencoder

Encoder & Decoder (MWCNN):

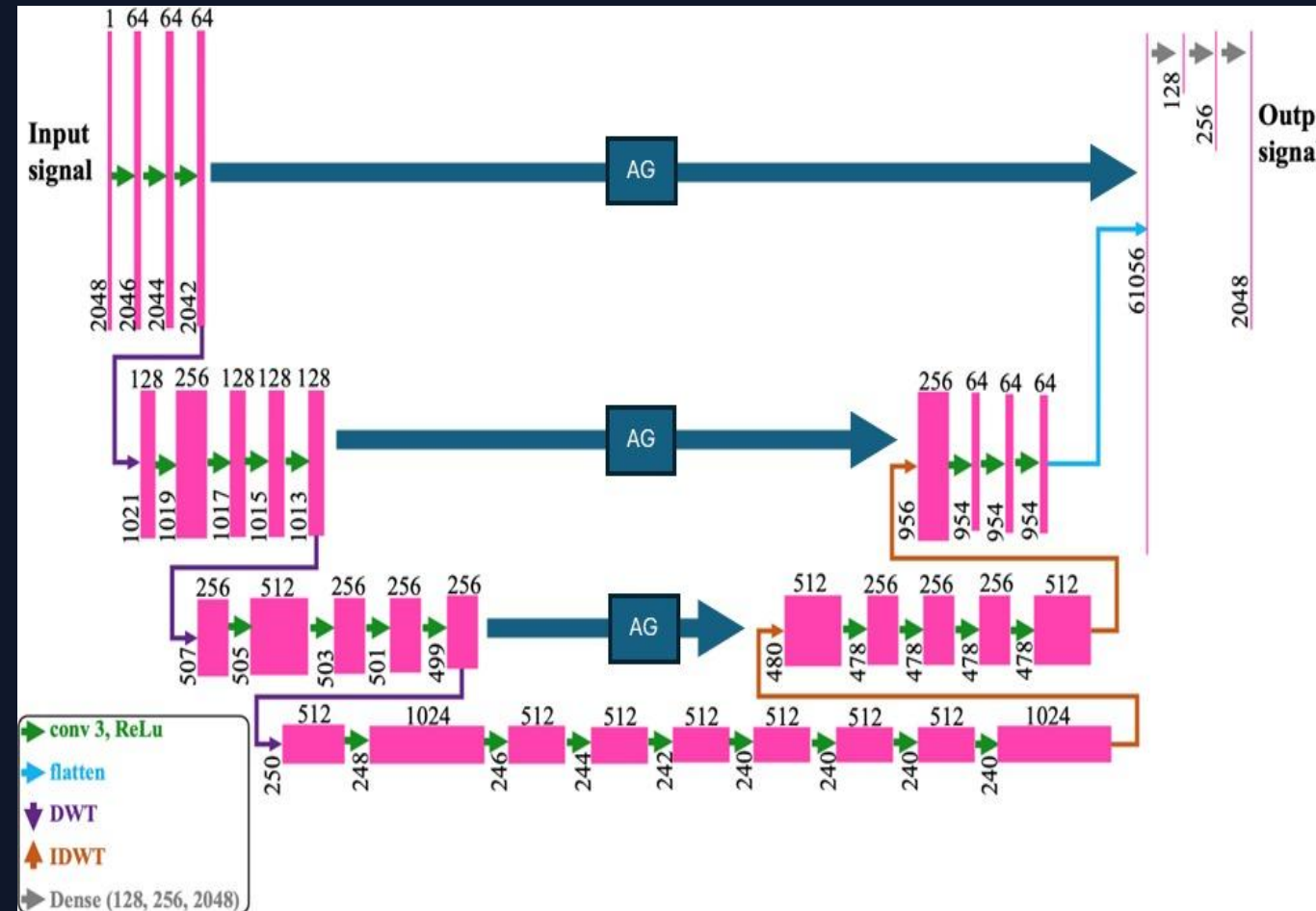
Replaces pooling and upsampling with Discrete Wavelet Transform (DWT) and IDWT for feature extraction and reconstruction.

Bottleneck:

Combines Swin Transformer (Global context) with BiGRU (Temporal dynamics).

Skip Connections:

Attention Gates actively filter noise before it reaches the decoder.



Baseline AI Models



PURE CNN

Standard Convolutional Neural Network. Good at local features but lacks global context.



1D U-NET

Encoder-Decoder with skip connections. Better restoration but standard pooling loses frequency details.



MWCNN

Multi-Level Wavelet CNN. Uses wavelets for downsampling, preserving frequency info.

Training Methodology



Multi-Task Learning

- Jointly trains for Denoising (Reconstruction) and Semantic Segmentation.
- The segmentation head acts as a regularizer to preserve pulse shape.



Curriculum Learning

- Training moves from Easy (High SNR) to Hard (Low SNR).
- Prevents convergence to local minima in extreme noise (-20dB).



Adaptive Loss

- Uses Charbonnier Loss for robustness and Focal Loss to handle class imbalance (sparse PD pulses).

Training Methodology



MSE / RMSE

Measures the pixel-level accuracy of the reconstruction.



SNR IMPROVEMENT

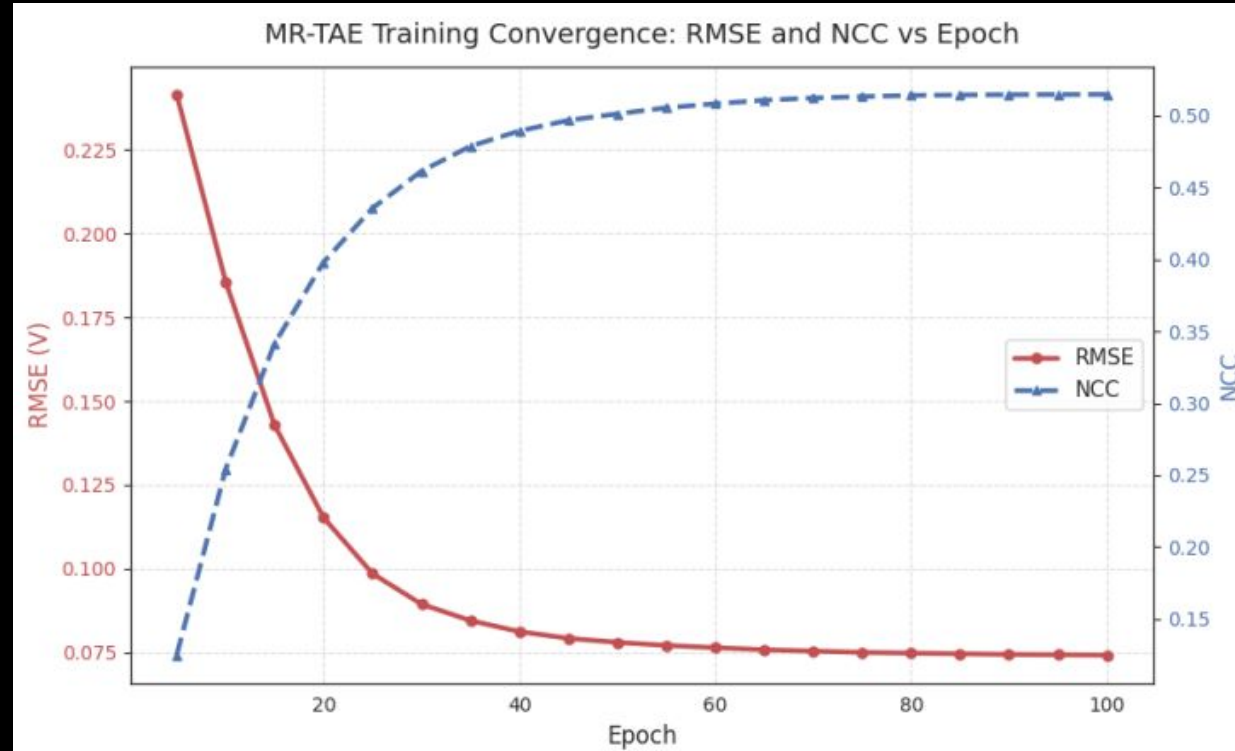
Measures noise removed relative to signal power.



NCC (CORRELATION)

Normalized Cross-Correlation.
Measures how well the **shape** was preserved.

Results



At Epoch 40, when curricular training enters phase 3 (Difficult Mode), the model still improves, proving the successful integration of Hybrid Bottleneck into the model

Results

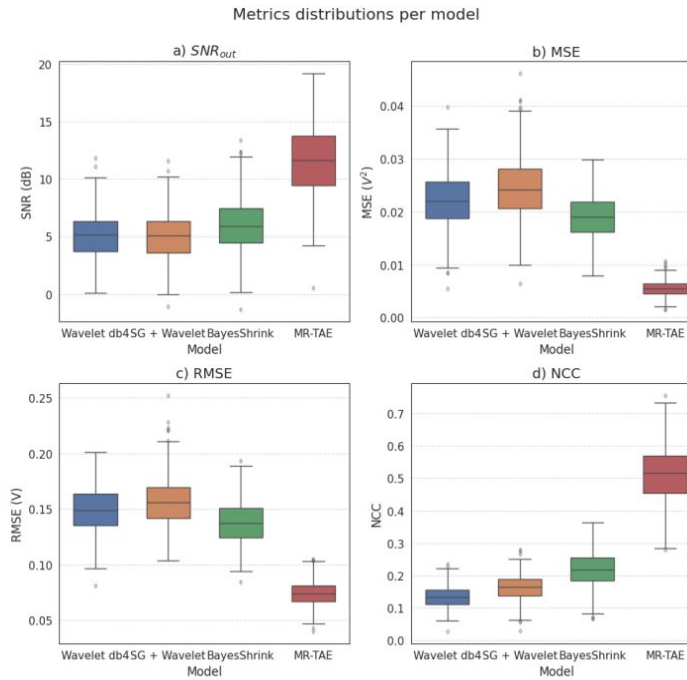
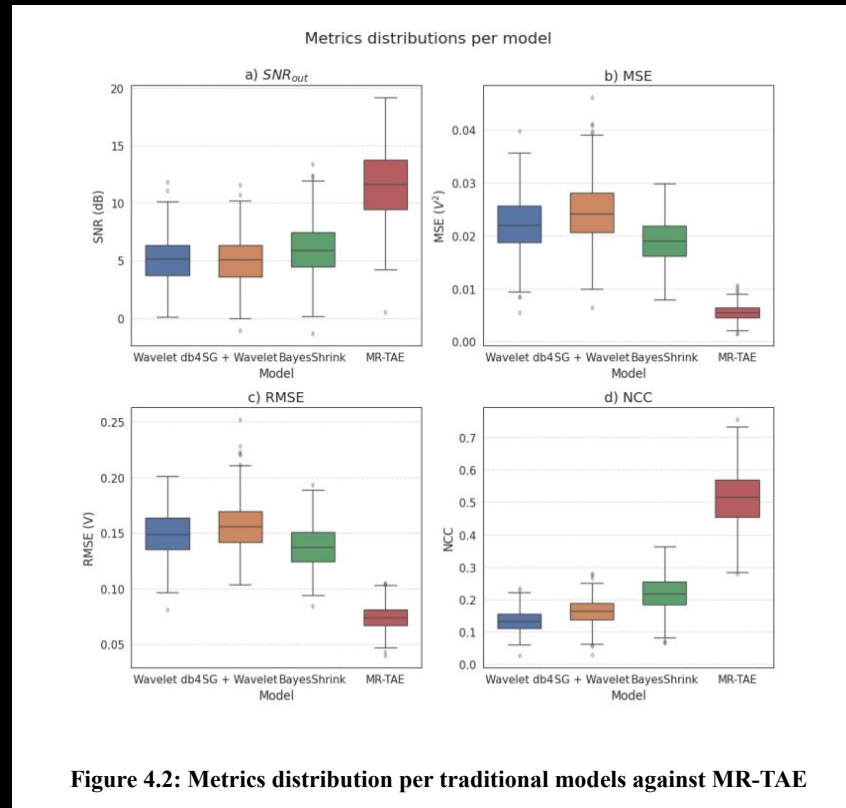


Figure 4.2: Metrics distribution per traditional models against MR-TAE

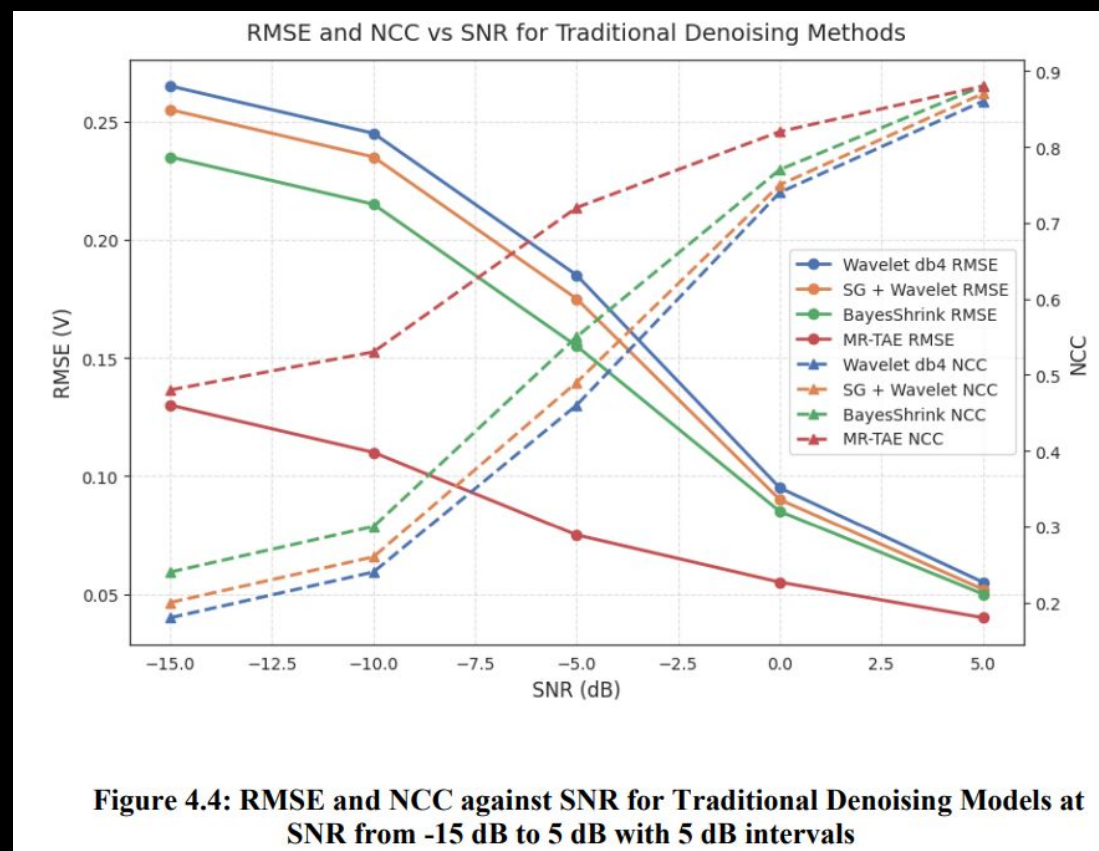
*Consistently demonstrate significantly better SNR and NCC,
lower MSE and RMSE, against AI Denoising Models*

Results



Consistently demonstrate significantly better SNR and NCC, lower MSE and RMSE, against traditional Denoising Models, MR-TAE exhibited a mean MSE of 0.0055 V^2 , representing a reduction of 71% compared to the BayesShrink(0.0189 V^2).

Results



MR-TAE demonstrated drastically better RMSE, and NCC result at very low SNR environment with remarkable stability with an RMSE of only 0.130 V, effectively half the error of the standard Wavelet method [0.235, 0.265].

Results: Signal-to-Noise Ratio



The MR-TAE achieved a massive +6.5 dB improvement over the best traditional method.

Diagnostic Integrity (Shape Fidelity)

0.51

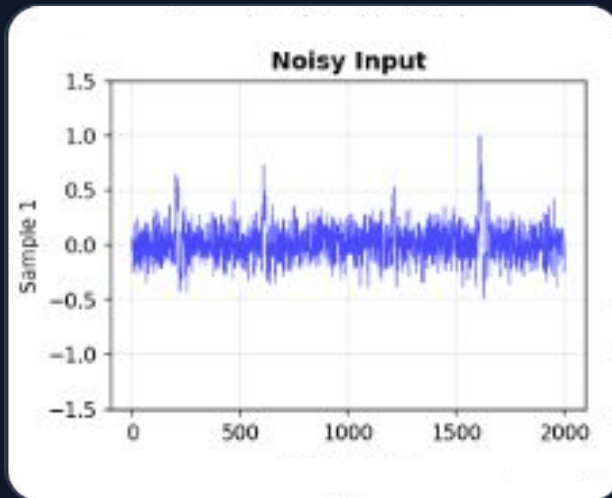
MR-TAE NCC Score
Normalized Cross-Correlation

0.22

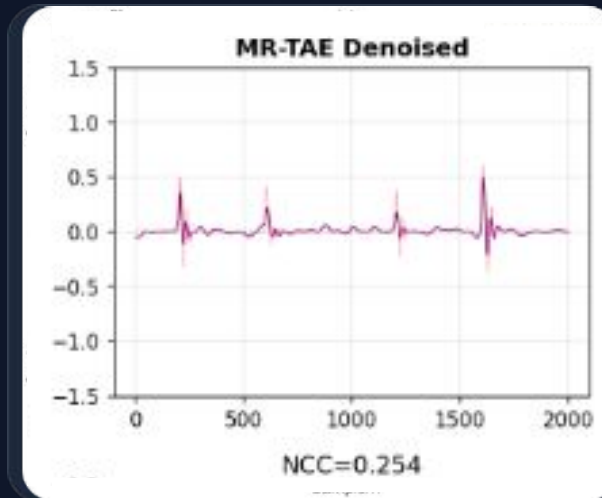
Traditional Methods
Average NCC Score

High NCC indicates the **morphological features** (rise time, decay) are preserved, which is critical for identifying the *type* of fault. Traditional methods blur these features.

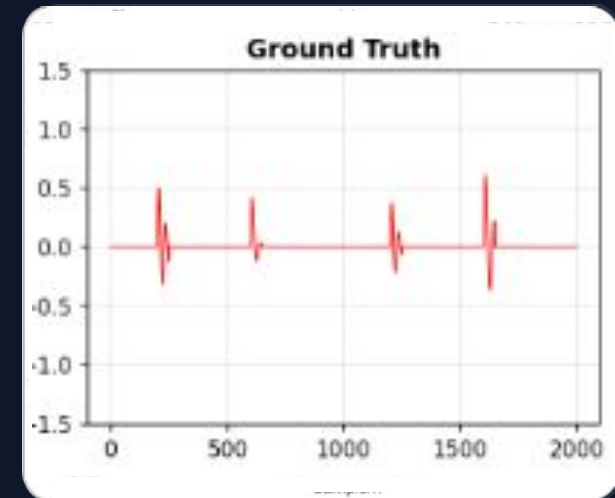
Visual Performance Demonstration



Noisy Input
Signal buried in noise

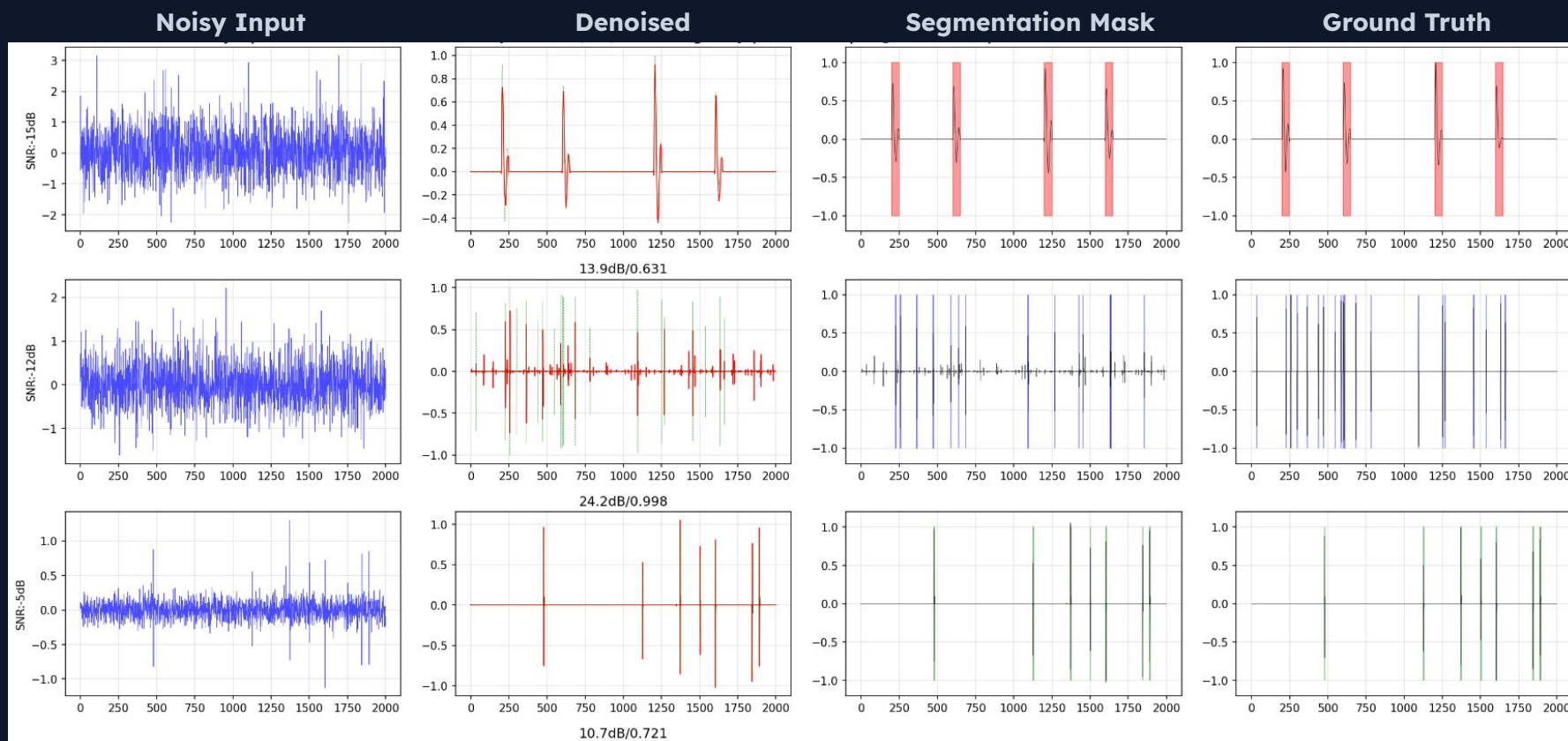


MR-TAE Output
Shape and Sharpness Preserved



Ground Truth
Original Clean Signal

Visual Performance Demonstration

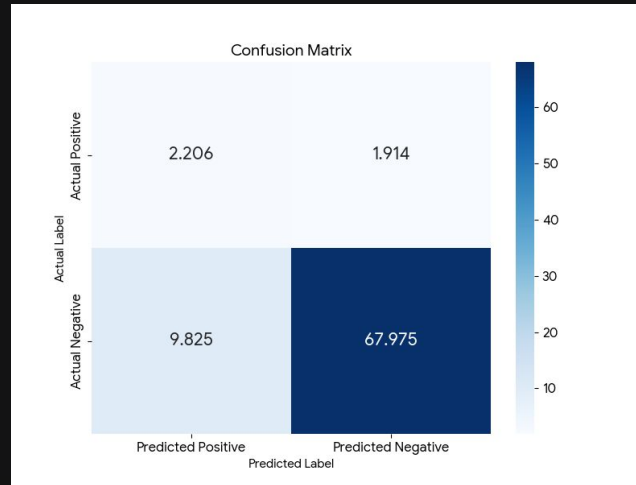


Classification Result

Type of Signal	Accuracy				
	Corona	Internal	Surface	PD	Non-PD
Clean Signal	1.355/1.365	1.266/1.275	1.470/1.480	4.091/4.120	0.050/77.80
Noisy Signal	0.247/1.365	0.255/1.275	0.082/1.480	0.584/4.120	45.50/77.80
Daubechies 4	0.651/1.365	0.353/1.275	0.259/1.480	1.263/4.120	28.63/77.80
BayersShrink	1,043/1.365	0.372/1.275	0.258/1.480	1.673/4.120	23.42/77.80
Savitzky-Golay	0.760/1.365	0.253/1.275	0.222/1.480	1.235/4.120	19.80/77.80
Pure CNN	0.852/1.365	0.343/1.275	0.299/1.480	1.494/4.120	16.51/77.80
1D U-Net	0.982/1.365	0.523/1.275	0.382/1.480	1.887/4.120	14.20/77.80
MWCNN	1.031/1.365	0.558/1.275	0.440/1.480	2.029/4.120	11.51/77.80
MR-TAE	1.125/1.365	0.598/1.275	0.483/1.480	2.206/4.120	9.825/77.80

Unit: Millions

Classification Result



$$\text{Precision} = \frac{TP}{TP + FP} = \frac{2.206}{2.206 + 9.825} = \frac{2.206}{12.031} \approx 0.1834$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2.206}{2.206 + 1.914} = \frac{2.206}{4.12} \approx 0.5354$$

$$\text{F1 Score} = 2 \times \frac{0.1834 \times 0.5354}{0.1834 + 0.5354} = 2 \times \frac{0.09819}{0.7188} \approx 0.2732$$

Conclusion

By replacing passive reconstruction with active filtering and joint optimization, MR-TAE provides a high-fidelity AI solution for real-world PD monitoring.